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This is a corrected copy of the paper cited above. The text is retyped from the printed paper. It keeps the paper's wording and the numbering of its results, equations and footnotes, and a marker such as [p. N] shows where page N of the printed paper begins.

The corrections come from a formalization of the Graph Minors series in the Lean proof assistant, github.com/mahdi-jfri/GraphMinor, in which the results of this paper are proved. Where the formalization found that a statement or a step of a proof is wrong as printed, this copy corrects it in place. Where the paper leaves a proof or a step to the reader, this copy adds it, also in blue. Text ~~struck through in red~~ is removed and text in blue is inserted. Each change has a note, marked F1, F2, ..., that gives the reason. The file corrections/GM06.md of the formalization describes each correction in full. Each note, and each item in the list below, begins with the kind of the change. A **misprint** is a wrong symbol, index or word. A **statement mistake** is a result or a claim of the text that is false as printed. A **degenerate case** is a statement that fails only in a trivial case, such as an empty graph or a parameter equal to 0. A **proof gap** is a step of a proof that does not follow as written. A **missing proof** is a proof that the paper omits, leaves to the reader or only sketches, and that this copy adds.

The corrections and added proofs, in the order of the pages:

1. **Missing proof.** Proof of (3.1), page 120: the proof is added.
2. **Missing proof.** Proof of (3.2), page 120: the proof, along the lines of the hint, is added.
3. **Missing proof.** Proof of (3.6), page 122: the proof of the necessity of (i) and (ii) is added.
4. **Proof gap.** Proof of (4.2), page 126: (4.1) and reduction 2 are applied only to consecutive vertices a, b of H' in S_i, for which $S_i(a, b)$ is a segment, and the conclusion is passed from consecutive pairs to all pairs.
5. **Proof gap.** Proof of (4.2), page 127: "distinct" becomes "consecutive", and the step to the connectivity of K is explained.

6. **Statement mistake.** Reduction 1 in Section 5, page 128: for $k = 1$, the deleted edge must leave s_1 on the side of the strip that precedes s_1.
7. **Statement mistake.** Reduction 4 in Section 5, page 129: for $k = 1$, the condition that v is incident with R_1 is read in the universal cover of the annulus.
8. **Missing proof.** Proof of (5.4), page 130: the details are added.
9. **Missing proof.** Algorithm (5.7), page 133: the count that bounds the running time is added.

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Graph Minors. VI. Disjoint Paths across a Disc

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Let G be a graph drawn on a disc, and let the vertices of G on the boundary of the disc be $s_1, \ldots, s_k, t_1, \ldots, t_k$ in some order. When are there k vertex-disjoint paths of G joining s_i and t_i $(1 \leq i \leq k)$, respectively? We give a structural characterization and a polynomial algorithm for this problem. We also solve the same question when the disc is replaced by a cylinder.

1. Introduction

The problem DISJOINT CONNECTING PATHS is, given a graph G and vertices $s_1, \ldots, s_k, t_1, \ldots, t_k$ of G, to decide if there are k vertex-disjoint paths of G between s_i and t_i $(1 \leq i \leq k)$, respectively. If k is a variable part of the input, this was shown to be NP-complete by Karp [1]; and Lynch [2] has shown that it remains NP-complete even if G is constrained to be planar. However, we suspect that if G is planar and $s_1, \ldots, s_k, t_1, \ldots, t_k$ are constrained to lie on some bounded number ρ of regions of G, the problem is polynomially solvable; and in this paper we show that this is true if $\rho = 1$ or 2.

We also conjecture that DISJOINT CONNECTING PATHS is polynomially solvable for general graphs G if k is fixed. This is true if $k \leq 2$, but is open for $k \geq 3$. We shall show in [3] that if k is fixed, DISJOINT CONNECTING PATHS is polynomially solvable if the genus of G is bounded. Indeed, that is one of the motivations for the present paper, for the results given here are basically lemmas for use in [3]. We are [p. 116] publishing them separately because we feel that they are of some independent interest.

It is important for our applications to deal with a slightly more general problem. (Conveniently, the generalization makes the problem essentially no harder.) A *division* is a finite set of non-empty pairwise disjoint sets. If Δ is a division, we abbreviate $\bigcup(\delta \in \Delta)$ by $\bigcup \Delta$. If G is a graph and Δ

is a division with $\bigcup \Delta \subseteq V(G)$, a *realization* of Δ in G is a subgraph of G which is a forest with $|\Delta|$ components $H(\delta)$ ($\delta \in \Delta$) such that $\delta \subseteq V(H(\delta))$ ($\delta \in \Delta$). We say that Δ is *feasible* in G if there is a realization of Δ in G. The problem DISJOINT CONNECTING SUBGRAPHS is, given a graph G and a division Δ, with $\bigcup \Delta$ a subset of $V(G)$, to decide if Δ is feasible in G. If each member of Δ has cardinality two this is clearly equivalent to DISJOINT CONNECTING PATHS. We shall give a polynomial algorithm for this problem when G is planar and there are two regions of G which together are incident with every vertex in $\bigcup \Delta$. Essentially equivalent to this and notationally more convenient is to constrain G to be drawn on a cylinder and $\bigcup \Delta$ to be a subset of the boundary of the cylinder.

Let us fix some terminology. A *surface* is a compact 2-manifold with boundary. In this paper we are only concerned with three surfaces, the disc, the Möbius band, and the cylinder, which is homeomorphic to

$$\{(x, y) : 1 \leq x^2 + y^2 \leq 4\}.$$

We generally denote a surface by Σ, and its boundary by $bd(\Sigma)$. Any component of $bd(\Sigma)$ is called a *cuff* of Σ. By a *drawing* of a graph G on Σ we shall always mean, as usual, a drawing without crossings. It will often be convenient to identify G with a drawing of G, and we do so when we expect it will cause no confusion. In particular, if G is drawn on Σ we sometimes regard G as a subset of Σ. A *region* of G is a connected component of $\Sigma - G$. In any drawing, every point of Σ is either a vertex of G, or lies in an edge, or lies in a region, and only one of these is true. A graph G is *properly drawn* on Σ if no edge of G meets $bd(\Sigma)$. (Thus, $bd(\Sigma)$ only contains vertices of G and points which are in regions.) Graphs may have loops or multiple edges. The *valency* of a vertex is the number of edges incident with it; loops are counted twice. We denote the closure of X by $\bar{X}$.

2. Divisions On A Circle

In the next section we shall give an algorithm and a structural characterization for our problem when G is drawn on a disc Σ and the division Δ has $\bigcup \Delta \subseteq bd(\Sigma)$. But first we develop some theory of divisions on a circle. (Our method for this disc case is rather more tortuous then necessary; a [p. 117] more direct approach is described in Section 4. We use our tortuous method because we find it entertaining, and we think the results we obtain along the way are worthwhile.)

Let C be homeomorphic to a circle and let $V \subseteq C$ be finite. Every component of $C - V$ homeomorphic to $(0, 1)$ is called a *segment* (of C with

respect to V). Now let Δ be a division with $\bigcup \Delta = V$. If $V = \emptyset$ let $U = \emptyset$. If $V \neq \emptyset$ let $U \subseteq C - V$ with $|U \cap S| = 1$ for every segment S of C with respect to V. Then $|U| = |V|$.

If a, b, c, d are distinct points of C occurring in that order, we say that $\{a, c\}$ *crosses* $\{b, d\}$. We define a division Δ^* with $\bigcup \Delta^* = U$ as follows. The relation between $u, u' \in U$, "there is no $\delta \in \Delta$ and $v, v' \in \delta$ such that $\{v, v'\}$ crosses $\{u, u'\}$" is an equivalence relation on U, as is easily seen. We denote the set of equivalence classes by Δ^*. We define Δ^{**} to be the set of equivalence classes on V of the relation between $v, v' \in V$, "there is no $\delta^* \in \Delta^*$ and $u, u' \in \delta^*$ such that $\{u, u'\}$ crosses $\{v, v'\}$."

It is not necessarily true that $\Delta^{**} = \Delta$. For example, if $V = \{v_1, v_2, v_3, v_4\}$ and v_1, v_2, v_3, v_4 occur on C in that order, and

$$\Delta = \{\{v_1, v_3\}, \{v_2, v_4\}\}$$

then Δ^* contains the singleton subsets of U, and Δ^{**} contains only the set V.

We say that Δ is *cross-free* if there do not exist distinct $\delta, \delta' \in \Delta$ and $s, t \in \delta$, $s', t' \in \delta'$ such that $\{s, t\}$ crosses $\{s', t'\}$. If Σ is a disc and $C = bd(\Sigma)$, and Δ is a division with $\bigcup \Delta \subseteq C$ feasible in some graph drawn in Σ, then it is easy to see that Δ must be cross-free.

(2.1) *Δ^* is cross-free, for any Δ.*

Proof. Let $u_1, u_2, u_3, u_4 \in U$ occur on C in that order, and suppose that

$$\{u_1, u_3\} \subseteq \delta_1^* \in \Delta^*;\ \{u_2, u_4\} \subseteq \delta_2^* \in \Delta^*.$$

We must show that $\delta_1^* = \delta_2^*$. Suppose not; then there exists $\delta \in \Delta$ and $v_1, v_2 \in \delta$ such that $\{v_1, v_2\}$ crosses $\{u_1, u_2\}$. But $\{v_1, v_2\}$ crosses neither $\{u_1, u_3\}$ nor $\{u_2, u_4\}$, and that is impossible. Hence $\delta_1^* = \delta_2^*$, as required.

(2.2) *If Δ is cross-free then $\Delta^{**} = \Delta$.*

Proof. Suppose that $v, v' \in V$ and there is no $\delta \in \Delta$ with $v, v' \in \delta$. We must show that there exists $\delta^* \in \Delta^*$ and $u, u' \in \delta^*$ such that $\{u, u'\}$ crosses [p. 118] $\{v, v'\}$. Let P_1, P_2 be the closures of the two components of $C - \{v, v'\}$. Choose $\delta \in \Delta$ with $v \in \delta$, and for $i = 1, 2$ choose $v_i \in P_i \cap \delta$ as close to v' as possible. (It is possible that $v_1, v_2 = v$.) Now $v' \notin \delta$, and so $v_1, v_2 \neq v'$. For $i = 1, 2$ let S_i be the segment of C with respect to V with $S_i \subseteq P_i$ and $v_i \in \bar{S}_i$, such that S_i and v' are in the same component of $C - \{v, v_i\}$. Let $U \cap S_i = \{u_i\}$ $(i = 1, 2)$. Now by choice of v_1, v_2 and since Δ is cross-free, there is no $\delta' \in \Delta$ and $w, w' \in \delta'$ such that $\{w, w'\}$ crosses $\{v_1, v_2\}$, and hence none such that $\{w, w'\}$ crosses $\{u_1, u_2\}$. Thus

there exists $\delta^* \in \Delta^*$ such that $u_1, u_2 \in \delta^*$. But $\{u_1, u_2\}$ crosses $\{v, v'\}$, as required.

Put $\Gamma = \Delta \cup \Delta^*$ and $W = V \cup U$.

(2.3) *If $W \neq \emptyset$, one of Δ, Δ^* has a member with cardinality* 1.

Proof. Choose $c \in C - W$, and let $W = \{w_1, \ldots, w_{2k}\}$, numbered in order on $C - \{c\}$. Choose i, j with $1 \leq i \leq j \leq 2k$ so that

(i) for some $\gamma \in \Gamma$, $\{x, c\}$ crosses $\{w_i, w_j\}$ for all $x \in \gamma - \{w_i, w_j\}$

(ii) subject to (i), $j - i$ is minimum.

(This is possible since the choice $i = 1$, $j = 2k$ satisfies (i), because $k \geq 1$.) If $j = i$ the theorem is true. Suppose for a contradiction that $j > i$, and choose γ satisfying (i). Now $w_i, w_j \in \gamma$ by the minimality of $j - i$, and so $j - i$ is even and hence $j - i \geq 2$. Choose $\gamma' \in \Gamma$ with $w_{i+1} \in \gamma'$. Now if $x \in \gamma'$ then $x \neq w_j$, and $\{w_{i+1}, x\}$ does not cross $\{w_i, w_j\}$ by definition of Δ^*; thus either $x = w_{i+1}$ or $\{x, c\}$ crosses $\{w_{i+1}, w_j\}$. This contradicts the minimality of $j - i$, as required.

For $x, y \in W$ we define

$$\Delta(x, y) = |\{\delta \in \Delta : \text{either } \delta \cap \{x, y\} \neq \emptyset, \text{ or there exist } u, v \in \delta \text{ such that } \{u, v\} \text{ crosses } \{x, y\}\}|,$$

and we define $\Delta^*(x, y)$ analogously.

(2.4) *If Δ is cross-free and $x, y \in W$, then*

$$\begin{aligned} \Delta(x, y) - \Delta^*(x, y) &= 1 && \text{if } x, y \in V \\ &= -1 && \text{if } x, y \in U. \\ &= 0 && \text{otherwise.} \end{aligned}$$

Proof. If $x = y$ the theorem may easily be verified, and we assume that $x \neq y$. Let C_1, C_2 be the closures of the two components of $C - \{x, y\}$. Let $v_1 \in C_1, v_2 \in C_2$. We say that (v_1, v_2) is a Δ-*minimizer* if

(i) $v_1, v_2 \in V$ and $v_1, v_2 \neq x$;

[p. 119] (ii) there exists $\delta \in \Delta$ such that $v_1, v_2 \in \delta$ and $\{v, x\}$ crosses $\{v_1, v_2\}$ for every $v \in \delta - \{v_1, v_2\}$.

It follows from (i) and (ii) that $x \notin \delta$. It is easy to see that for any $\delta \in \Delta$, if $\delta \cap C_1 \neq \emptyset$, $\delta \cap C_2 \neq \emptyset$ and $x \notin \delta$ then there is a unique Δ-minimizer (v_1, v_2) with $v_1, v_2 \in \delta$. Thus the number of Δ-minimizers equals

$$\Delta(x, y) - |V \cap \{x\}|.$$

Again, let $u_1 \in C_1$, $u_2 \in C_2$. We say that (u_1, u_2) is a Δ^*-*maximizer* if

(i) $u_1, u_2 \in U$ and $u_1, u_2 \neq y$;

(ii) there exists $\delta^* \in \Delta^*$ such that $u_1, u_2 \in \delta^*$ and $\{u, y\}$ crosses $\{u_1, u_2\}$ for every $u \in \delta^* - \{u_1, u_2\}$.

Then similarly the number of Δ^*-maximizers equals

$$\Delta^*(x, y) - |U \cap \{y\}|.$$

On the other hand, there is a 1-1 correspondence between Δ-minimizers and Δ^*-maximizers. For let (v_1, v_2) be a Δ-minimizer. Let $u_i \in U \cap C_i$ be chosen to lie in C_i between v_i and x, as close to v_i as possible $(i = 1, 2)$. It is easy to see that (u_1, u_2) is a Δ^*-maximizer, and is uniquely given by (v_1, v_2). Moreover, by (2.2) the construction may be reversed, to obtain a Δ-minimizer from any Δ^*-maximizer, and we have a 1-1 correspondence.

Putting these results together, we obtain

$$\Delta(x, y) - |V \cap \{x\}| = \Delta^*(x, y) - |U \cap \{y\}|,$$

which proves the theorem.

3. The Disjoint Subgraph Problem on a Disc

Let G be a connected graph, properly drawn on a (closed) disc Σ. We define the *dual graph* G^* to have vertex set the set of all regions of G, and to have edges in 1-1 correspondence with the edges of G; if e is an edge of G and R_1, R_2 are the regions of G on either side of e then the corresponding edge of G^* has ends R_1, R_2. We may obtain a drawing of G^* in Σ as usual by choosing a point from each region of G to represent vertices of G^*, and for each edge of G drawing the corresponding edge of G^* crossing it in the usual way. We may further arrange that

(i) G^* is properly drawn in Σ, and

(ii) each segment (of $bd(\Sigma)$ with respect to $V(G) \cap bd(\Sigma)$) contains a vertex of G^*, and no other vertices of G^* lie in $bd(\Sigma)$.

[p. 120] Here (ii) is possible since any region of G contains at most one segment, because G is connected. Henceforth by G^* we shall mean not only the abstract dual graph but also a drawing of it satisfying the above conditions.

Standard arguments from duality theory for graphs on the sphere may be used to obtain the following lemma, whose proof is ~~left to the reader~~given below.

(3.1) *If G is connected and properly drawn on a disc Σ, then $G^{**} = G$. (More precisely, the drawing of G is a drawing of G^{**}.)*

We give the argument. We read "in the usual way" as follows. Each region of G contains exactly one vertex of G^*, and each vertex of G^* lies in a region of G. For each edge e of G, the corresponding edge e^* of G^* meets the drawing of G only in points of e, the edge e meets the drawing of G^* only in points of e^*, and e, e^* have exactly one point in common. The ends of e^* are the vertices of G^* in the regions of G incident with e, and each region of G^* incident with e^* contains a point of e. Finally, distinct vertices of G lie in distinct regions of G^*. Then (3.1) says that the drawing of G satisfies these conditions with the roles of G and G^* exchanged, and also condition (ii) above with the roles exchanged.

No vertex of G lies in the drawing of G^*, since the vertices of G^* lie in regions of G, and each edge e^* meets the drawing of G only in points of e. So each vertex of G lies in a region of G^*. Next let e be an edge of G with ends u, v. Removing from e its point in e^* leaves two connected pieces, one with u and one with v in its closure. They are disjoint from the drawing of G^*, and so they lie in the regions of G^* containing u and v respectively. Hence these two regions are incident with e^*, since they contain points of e arbitrarily close to the point in e^*. They are the only regions of G^* incident with e^*, since every such region contains a point of e.

Now let r be a region of G^*. Then r contains at most one vertex of G, since distinct vertices of G lie in distinct regions of G^*. If r is incident with some edge e^*, then r contains an end of e, by the previous paragraph. If r is incident with no edge of G^*, then every point of the frontier of r is a vertex of G^*. Since Σ minus finitely many points is connected, it follows that r includes Σ minus the vertices of G^*, and so r contains every vertex of G. There is such a vertex, since G is connected. Thus each region of G^* contains exactly one vertex of G. Next, if $V(G) \cap bd(\Sigma)$ is non-empty, then by (ii) above its points alternate around $bd(\Sigma)$ with those of $V(G^*) \cap bd(\Sigma)$, and so each segment of $bd(\Sigma)$ with respect to $V(G^*) \cap bd(\Sigma)$ contains exactly one vertex of G. If $V(G) \cap bd(\Sigma)$ is empty, then so is $V(G^*) \cap bd(\Sigma)$, since there are no segments. Next, distinct vertices of G^* lie in distinct regions of G, since each region of G contains only one vertex of G^*. Finally, let e be an edge of G. The conditions on the points of e and e^* are symmetric in e and e^*. The ends of e are the vertices of G in the regions of G^* incident with e^*, by the second paragraph. Each region of G incident with e contains an end of e^*, and so contains a point of e^*, since it is open and the end is in the closure of e^*. This proves (3.1).[F1]

[F1] **Missing proof.** The paper leaves (3.1) to the reader. The argument above is the one used in the formalization, which reads "in the usual way" by the conditions stated at its

Now we return to the Disjoint Connecting Subgraphs problem. Let G be a graph drawn on a disc Σ, and let Δ be a division with

$$\bigcup \Delta \subseteq V(G) \cap bd(\Sigma).$$

We are concerned with deciding if Δ is feasible in G. By slightly enlarging Σ we may arrange that

(i) G is properly drawn on Σ, and

(ii) $\bigcup \Delta = V(G) \cap bd(\Sigma)$

and for the rest of this section we shall assume that (i) and (ii) hold.

(3.2) *Let G be a connected graph, properly drawn on a disc Σ. Let Δ be a cross-free division with*

$$\bigcup \Delta = V(G) \cap bd(\Sigma).$$

Let the drawing of G^* be chosen so that

$$\bigcup \Delta^* = V(G^*) \cap bd(\Sigma).$$

Then Δ is feasible in G if and only if Δ^* is feasible in G^*.

This can be proved using standard duality arguments~~, and is left to the reader~~. (Hint: a realization of Δ^* in G^* may be obtained from some of the edges of G^* corresponding to edges of G not used by some given realization of Δ in G.)

We give the argument, which follows the hint. Let $U = V(G^*) \cap bd(\Sigma)$. Suppose first that Δ is feasible in G, and let H be a realization of Δ in G. Let K^* be the subgraph of G^* with vertex set $V(G^*)$ whose edges are the edges e^* for the edges e of G not in H. We claim that for $u_1, u_2 \in U$, there is a path of K^* between u_1 and u_2 if and only if u_1 and u_2 belong to the same member of Δ^*.

Suppose that there is such a path P, and that $\{v_1, v_2\}$ crosses $\{u_1, u_2\}$ for some $\delta \in \Delta$ and $v_1, v_2 \in \delta$. Let Q be the path of $H(\delta)$ between v_1 and v_2. The drawings of P and Q are compact connected subsets of Σ, and their ends on $bd(\Sigma)$ cross, and so they meet. Let z be a common point. Then z is in the drawing of G, and so it is not a vertex of G^*, since the vertices of G^* lie in regions of G. So z lies in an edge f^* of K^*. Since f^* meets the drawing of G only in points of f, the point z lies in f. But f meets no vertex of G and no other edge of G, and f is not an edge of H, and so z is

start.

not in Q. This is a contradiction, and so u_1, u_2 belong to the same member of Δ^*.

Conversely, suppose that they do. Let Γ be the drawing of the subgraph of G with vertex set $V(G)$ and edge set $E(H)$. We show first that no component of Γ has two vertices v_1, v_2 in $bd(\Sigma)$ such that $\{v_1, v_2\}$ crosses $\{u_1, u_2\}$. Such vertices are in $\bigcup \Delta$, which is a subset of $V(H)$, and they are joined by a path of H. So they lie in the same component $H(\delta)$ of H, and hence both belong to δ, since every other member of Δ is included in another component of H. Since u_1, u_2 are in the same member of Δ^*, it follows that $\{v_1, v_2\}$ does not cross $\{u_1, u_2\}$. Next we show that u_1 and u_2 lie in the same region of Γ. Let B_1, B_2 be the two arcs of $bd(\Sigma)$ with ends u_1, u_2. Let K_1 be the union of the drawings of the components of Γ whose vertices in $bd(\Sigma)$ all lie in B_2, and let K_2 be the union of the drawings of the other components. Then K_1, K_2 are disjoint closed sets with union the drawing of Γ. The arc B_1 is disjoint from K_1, since u_1, u_2 are not in the drawing of Γ. The arc B_2 is disjoint from K_2, since a component in K_2 has a vertex in $bd(\Sigma) - B_2$, and a vertex in B_2 would make a pair that crosses $\{u_1, u_2\}$. By Janiszewski's theorem, applied in the disc to K_1 and K_2, the points u_1, u_2 lie in the same component of Σ minus $K_1 \cup K_2$, that is, in the same region r of Γ. Now let Z be the set of regions of G whose vertex of G^* is joined to u_1 by a path of K^*, and let r_1, r_2 be the regions of G containing u_1, u_2. Both are included in r, since they are connected and disjoint from the drawing of Γ, and $r_1 \in Z$. Each point of r lies in a region of G included in r, or in an edge of G not in H, and such an edge lies in r together with every region of G incident with it. Since r is connected, it follows that if $r_2 \notin Z$, then some edge e of G not in H is incident with a region in Z and a region not in Z. But then e^* is an edge of K^* joining the vertices of G^* in these two regions, which is impossible. Hence $r_2 \in Z$, and since u_2 is the vertex of G^* in r_2, this proves the claim.

By the claim, each member of Δ^* is included in the vertex set of a component of K^*, and distinct members of Δ^* are included in distinct components. Choosing a spanning tree in each of these components gives a realization of Δ^* in G^*. So Δ^* is feasible in G^*.

Now suppose that Δ^* is feasible in G^*. Since G is connected, (3.1) applies, and so the drawing of G is a drawing of the dual of G^*, and $V(G) \cap bd(\Sigma)$ contains exactly one point of each segment of $bd(\Sigma)$ with respect to U. Hence what we just proved applies to G^* and Δ^* in place of G and Δ, and shows that Δ^{**} is feasible in G. Since Δ is cross-free, $\Delta^{**} = \Delta$ by (2.2),

and so Δ is feasible in G.[F2]

We shall make use of the following observations.

(3.3) (i) *If $|\delta| = 1$ for some $\delta \in \Delta$, and $\delta = \{v\}$ say, then Δ is feasible in G if and only if $\Delta - \{\delta\}$ is feasible in the graph obtained from G by deleting v.*

(ii) *If G has a component G' with no vertex in $bd(\Sigma)$ then Δ is feasible in G if and only if it is feasible in the graph obtained from G by deleting $V(G')$.*

[p. 121] (iii) *If there exists $\delta \in \Delta$ which intersects more than one component of G then Δ is not feasible in G.*

(iv) *If G has components $G_1, \ldots, G_k$ for $k \geq 2$, and (ii), (iii) do not apply, let $(\Delta_1, \ldots, \Delta_k)$ be the partition of Δ such that*

$$\bigcup \Delta_i = V(G_i) \cap bd(\Sigma) \qquad (1 \leq i \leq k);$$

then Δ is feasible in G if and only if Δ_i is feasible in G_i for each i.

The proofs are all obvious. These observations, together with (3.2) and (2.3), yield a polynomially bounded algorithm, as follows.

(3.4) *Algorithm to decide if Δ is feasible in G.*

We first check that Δ is cross-free. (If not it is not feasible.) We apply (3.3) as often as possible, reducing the problem on G to equivalent problems on smaller graphs. If at any step (3.3)(iii) applies then Δ is not feasible in G, and we stop. If (3.3) cannot be applied and G is non-null then G is connected, and we may pass to G^*, Δ^* and so by (2.3) we may continue the reductions. Each reduction decreases the number of vertices plus the number of regions of G.

It is of course not necessary to use the dual graph G^*; we have done so merely for ease of exposition. In practise it would be easier to apply a dual form of (3.3)(i) to G, rather than apply (3.3)(i) to G^*. But this involves some messy technicalities which we prefer to avoid.

We turn now to the structural characterization for non-feasibility. Let G, Σ, Δ be as before. A vertex v of G is *peripheral* if $v \in bd(\Sigma)$, and a region is *peripheral* if it contains a segment of $bd(\Sigma)$ (with respect to $\bigcup \Delta$). Let $U \subseteq bd(\Sigma) - \bigcup \Delta$, such that $|U \cap S| = 1$ for each segment S and define Δ^* with $\bigcup \Delta^* = U$, as before. Put $W = U \cup (\bigcup \Delta)$. For $x, y \in W$, an *(x, y)-chain* is a sequence $A_1, A_2, \ldots, A_k$ with $k \geq 1$ such that

[F2] **Missing proof.** The paper leaves (3.2) to the reader, with the hint given. The argument above is the one used in the formalization, which uses the hint.

(i) for $1 \leq i < k$, one of A_i, A_{i+1} is a vertex of G, the other is a region, and they are incident

(ii) if A_1 is a vertex then $A_1 = x$; if A_1 is a region then $x \in A_1$; if A_k is a vertex then $A_k = y$; if A_k is a region then $y \in A_k$

(iii) for $1 \leq i \leq k$, A_i is peripheral if and only if $i = 1$ or $i = k$.

The *length* of an (x, y)-chain is the number of terms which are vertices. The Δ-*redundancy* of an (x, y)-chain in G is its length minus $\Delta(x, y)$.

(3.5) *If G is connected, any (x, y)-chain of G^* corresponds to an (x, y)-chain of G, and if Δ is cross-free the Δ^*-redundancy of the first chain equals the Δ-redundancy of the second.*

[p. 122] This follows easily from (2.4). Let us now prove our characterization.

(3.6) *Let G be properly drawn on a disc Σ, and let Δ be a division with*

$$\bigcup \Delta = V(G) \cap bd(\Sigma).$$

Then Δ is feasible in G if and only if (i) *Δ is cross-free, and* (ii) *every (x, y)-chain has non-negative redundancy, for all $x, y \in W$ (defined as before).*

Proof. The necessity of (i) and (ii) is easy. To see this, let H be a realization of Δ in G. We use the fact that two compact connected subsets of the disc Σ must meet if they contain pairs of points of $bd(\Sigma)$ that cross. If $a, c \in \delta$ and $b, d \in \delta'$ for distinct $\delta, \delta' \in \Delta$, and $\{a, c\}$ crosses $\{b, d\}$, then by this fact the drawings of a path of $H(\delta)$ between a and c and a path of $H(\delta')$ between b and d meet. That is impossible, since $H(\delta)$ and $H(\delta')$ are disjoint subgraphs of G. So (i) holds. For (ii), let $A_1, \ldots, A_k$ be an (x, y)-chain. For each term A_i that is a region, choose a curve in Σ whose interior lies in A_i, from x if $i = 1$ and from A_{i-1} if $i > 1$, to y if $i = k$ and to A_{i+1} if $i < k$. This is possible since the two ends lie in the closure of A_i, and any two points in the closure of a region are joined by such a curve. Let F be the union of these curves and of the terms that are vertices. Then F is compact and connected, it contains x and y, and it meets the drawing of G only in the terms that are vertices. Now let $\delta \in \Delta$ be one of the members counted in $\Delta(x, y)$. If $x \in \delta$, then x is a vertex of G and so lies in no region, and hence $A_1 = x$ by condition (ii) in the definition of a chain. Similarly, if $y \in \delta$ then $A_k = y$. If there exist $u, v \in \delta$ such that $\{u, v\}$ crosses $\{x, y\}$, then the drawing of a path of $H(\delta)$ between u and v meets F, by the fact above, and the common point is a term of the chain. In each case some term of the chain that is a vertex belongs to $H(\delta)$. Since the subgraphs $H(\delta)$ ($\delta \in \Delta$) are pairwise disjoint, distinct members of Δ

give distinct terms. Hence $\Delta(x, y)$ is at most the length of the chain, and so (ii) holds.[F3]

We prove their sufficiency by induction on the number of vertices of G plus the number of regions of G. If G is null the result is true. If G, Δ satisfy (3.6)(i) and (ii), and (3.3)(i), (ii) or (iv) applies, the smaller structures produced by this application also satisfy (3.6)(i) and (ii). Hence, by our inductive hypothesis, these smaller divisions are feasible in the corresponding graphs, and hence Δ is feasible in G. (To see this in the case of (3.3)(i), we observe that any (x, y)-chain of the graph obtained from G by deleting v which does not directly give a chain of G may be converted to a chain of G by adding v in the natural way. The resulting chain has length one more than the original, and has redundancy the same as the original.) If (3.3)(iii) applies, there is an (x, y)-chain consisting of a single region and no vertices where $\Delta(x, y) > 0$, contrary to (3.6)(ii), which is impossible. If none of (3.3)(i), ..., (iv) applies, and G is non-null then G is connected and we may pass to G^*, Δ^*. Now Δ^* is cross-free by (2.1), and every (x, y)-chain in G^* has non-negative Δ^*-reduncancy, by (3.5); thus G^*, Δ^* satisfy (3.6)(i) and (ii). Moreover (3.3)(i) applies to G^*, Δ^* by (2.3), and so Δ^* is feasible in G^*. Hence by (3.2), Δ is feasible in G. This completes the proof.

4. The Cylinder

Our object in this section and the next two is to extend our results about the Disjoint Connecting Subgraphs problem from the disc to the cylinder, where again we assume that $\bigcup \Delta$ is a subset of $bd(\Sigma)$.

A fundamental difference between the disc and cylinder problems is that in the case of the cylinder there is more than one homotopically inequivalent route between pairs of points of $bd(\Sigma)$, even if we restrict ourselves to non-self-intersecting routes, which we can. If s, t lie on the same cuff of the cylinder, there are in fact two homotopy classes of non-self-intersecting routes between s and t, while if s and t lie on opposite cuffs, there are infinitely many such routes, winding different numbers of times around [p. 123] the cylinder. We thus have a decision to make; for a natural variant of say DISJOINT CONNECTING PATHS on a cylinder is to specify not only the ends of each path but its homotopy class as well. Trial has convinced us that this variant is worthwhile—for the corresponding algorithm is simpler, and we can use it to solve the original DISJOINT CONNECTING PATHS problem on the cylinder. Analogous remarks apply for DISJOINT

[F3] **Missing proof.** The paper says that the necessity of (i) and (ii) is easy and gives no proof. The argument above is the one used in the formalization.

CONNECTING SUBGRAPHS, with a suitable definition of homotopy. We begin by making such a definition.

Let Σ be a surface. Let $s, t \in bd(\Sigma)$, and let $f_i\colon [0,1] \to \Sigma$ be continuous, with $f_i(0) = s$, $f_i(1) = t$ $(i = 1, 2)$. We say that f_1 is homotopic to f_2 if there is a continuous function $f\colon [1,2] \times [0,1] \to \Sigma$ such that

$$\begin{aligned} f(1,y) &= f_1(y) && (0 \le y \le 1)\\ f(2,y) &= f_2(y) && (0 \le y \le 1)\\ f(x,0) &= s && (1 \le x \le 2)\\ f(x,1) &= t && (1 \le x \le 2). \end{aligned}$$

Now let P_1, P_2 be graphs drawn on Σ, and suppose that P_1, P_2 are both paths, and both with end vertices $s, t \in bd(\Sigma)$. Let $f_i\colon [0,1] \to \Sigma$ be a continuous injection with image P_i and $f_i(0) = s$, $f_i(1) = t$ $(i = 1, 2)$. We say that P_1 is homotopic to P_2 if f_1 is homotopic to f_2. It is easy to see that this definition does not depend on the choice of f_1 and f_2.

Finally, let H_1, H_2 be forests drawn in Σ. We say that H_1 is homotopic to H_2 if

(i) $V(H_1) \cap bd(\Sigma) = V(H_2) \cap bd(\Sigma) = V$ say;

(ii) for $s, t \in V$, there is a path of H_1 between s and t if and only if there is a path of H_2 between s and t; and if these paths exist, one is homotopic to the other.

Our variant of DISJOINT CONNECTING SUBGRAPHS is the following.

ROUTED FOREST CONTAINMENT (ON Σ)

INSTANCE A surface Σ, a graph G drawn on Σ, and a forest H drawn on Σ with $V(H) \cap bd(\Sigma) \subseteq V(G)$.

QUESTION Does G have a forest subgraph homotopic to H?

Our approach to this is in two parts. We first give some reductions in the spirit of (3.3), which when Σ is a cylinder serve to reduce to a simpler problem, viz. when each component of H has exactly one vertex in each cuff of the cylinder. The simpler problem is solved in the next section, and in the remainder of this section we describe the reductions.

[p. 124] For convenience, we assume that G is properly drawn on Σ, and

$$V(H) \cap bd(\Sigma) = V(G) \cap bd(\Sigma).$$

(As before, this can always be arranged by extending Σ slightly.) Until further notice, Σ is a general surface, not necessarily a cylinder.

Reduction 1. If some component H' of H is such that G has a subgraph with vertex set $V(H') \cap bd(\Sigma)$ which is homotopic to H', then by deleting

$V(H')$ from H and deleting $V(H') \cap bd(\Sigma)$ from G, we obtain an equivalent problem.

This is easy to see.

Suppose that C is a cuff of Σ, and $s, t \in C$ are distinct, and S is a component of $C - \{s, t\}$. Suppose that P is a path drawn on Σ with end-vertices s and t. Then we define "P is homotopic to S" in the obvious way.

Again, if C, s, t, S are as above, and G is drawn on Σ, and $s, t \in V(G)$, and there is a path P of G homotopic to S, then there is such a path P of G where the portion of Σ enclosed by $P \cup S$ is minimal, and this path is unique. (A path has no "repeated" vertices.) This path is denoted by $P(s, t, S)$.

A *segment* is a component of $bd(\Sigma) - (V(H) \cap bd(\Sigma))$ homeomorphic to $(0, 1)$ and $v \in V(H) \cap bd(\Sigma)$ is an *end* of segment S if v is in the closure of S. The following describes some circumstances when the problem is infeasible.

(4.1) *With Σ, G, H as before, if there is a segment S with distinct ends s, t, and s, t are joined in H by a path homotopic to S, and if either*

(i) *$P(s, t, S)$ does not exist, or*

(ii) *some vertex of $P(s, t, S)$ lies in $V(H') \cap bd(\Sigma)$ for some component H' of H not containing s, t, or*

(iii) *some vertex v of $P(s, t, S)$ lies in $bd(\Sigma)$, and is joined to t in H by a path not homotopic to the subpath of $P(s, t, S)$ between v and t*

then G has no forest subgraph homotopic to H.

The proof is clear. We shall also require a further reduction.

Reduction 2. If there is a segment S with distinct ends s, t, and s, t are joined in H by a path homotopic to S, and

(i) *if $P(s, t, S) \cup S$ properly encloses a vertex or edge of G we may delete that vertex or edge;*

(ii) *if $P(s, t, S)$ has a vertex not in $bd(\Sigma)$, we may contract in G either edge of $P(s, t, S)$ incident with it;*

(iii) *if (i) above does not apply and $P(s, t, S)$ consists of a single edge e say, then we may contract e in G (moving s along S to t), make the iden-* [p. 125] *tification $s = t$ in H (again, moving along S to t) and delete one edge of the circuit thus formed in H.*

In each case the smaller problem is equivalent to the original.

Again, this is easy to see. What is less clear is what to do if none of these apply. One can show that if Σ is a disc then one of (4.1), reduction 1 and reduction 2 will apply, and we thus obtain another form of the algorithm for the disc; and indeed the characterization (3.6) can be proved this way

as well. For the cylinder it is possible that none of them apply.

(4.2) *If Σ is a cylinder and neither (4.1) nor reduction 1 nor reduction 2 applies, then each component of H has exactly one vertex in each cuff of Σ.*

For the proof we need a lemma. Let C be a cuff of a cylinder Σ, and let $S \subset C$ be connected. For distinct $a, b \in S$ we denote by $S(a, b)$ the component of $C - \{a, b\}$ which is contained in S.

(4.3) *Let Σ be a cylinder, and let C be a cuff of Σ. Let $s, t \in C$ be distinct, and let S be a component of $C - \{s, t\}$. Let K be a connected graph with $|V(K)| \geq 2$, properly drawn on Σ with $V(K) \subseteq S$. Then there is an edge of K with distinct ends a, b say, such that $S(a, b) \cap V(K) = \emptyset$.*

Proof. Choose distinct $a, b \in S$, adjacent in K, so that $S(a, b)$ is minimal. Suppose that $S(a, b) \cap V(K) \neq \emptyset$, and choose

$$c \in S(a, b) \cap V(K).$$

Since K is connected, there is a vertex $d \neq c$ adjacent in K to c, and then $d \in S(a, b) \cup \{a, b\}$. Hence $S(c, d) \subset S(a, b)$, a contradiction as required.

Proof of (4.2). No component of H has fewer than two vertices in $bd(\Sigma)$, because reduction 1 cannot be applied. Let us say a component of H is *bad* if it has more than one vertex on some cuff of Σ. Then we must prove that no component of H is bad. For distinct $a, b \in V(H) \cap bd(\Sigma)$, in the same component of H, let $H(a, b)$ denote the path of H between a and b.

(1) *Let C be a cuff of Σ, and let $a, b \in C \cap V(H)$ be distinct. Suppose that $H(a, b)$ exists, and let S be the component of $C - \{a, b\}$ homotopic to $H(a, b)$. Then for all $c, d \in S \cup \{a, b\}$, if $H(c, d)$ exists then it is homotopic to $S(c, d)$.*

For if (1) is not true then $H(a, b) \cup H(c, d)$ includes a circuit, which is impossible since H is a forest.

Let C be a cuff of Σ, and let H' be a component of H. Let $s, t \in C - V(H)$ [p. 126] be distinct, and let S be a component of $C - \{s, t\}$. We say that S is a C-base for H' if

$$V(H') \cap C = V(H) \cap S$$

and if for all distinct $a, b \in V(H) \cap S$, $H(a, b)$ is homotopic to $S(a, b)$. (We permit $V(H) \cap S = \emptyset$.) If H' has a C-base we say that H' is *interval-based on C*.

(2) *If some component of H is bad, there is a bad component of H which is interval-based on both cuffs of Σ.*

For if each component of H meets both cuffs of Σ then clearly each component is interval-based on both cuffs of Σ, and so in this case any bad

component satisfies (2). If some component meets only one cuff C say, then we may choose distinct $s, t \in C - V(H)$ and a component S of $C - \{s, t\}$ and a component H' of H, such that

(i) $V(H') \cap bd(\Sigma) \subseteq S$

(ii) for all distinct $a, b \in V(H') \cap S$, $H(a, b)$ is homotopic to $S(a, b)$.

(For since H' is a tree, there is a region R of H' which intersects both cuffs of Σ; choose $s, t \in R \cap C$.) Let us choose H', s, t, S, so that in addition $|S \cap V(H)|$ is minimum. It is easy to see, using (1), that S is a C-base for H'. Thus H' is interval-based on both cuffs of Σ, and it is clearly bad (since H' only meets one cuff of Σ). This proves (2).

Now to complete the proof of (4.2), suppose for a contradiction that H has a bad component. By (2) there is a bad component H' of H, interval-based on both cuffs of Σ. Let the cuffs of Σ be C_1, C_2, and let S_i be a C_i-base for H' $(i = 1, 2)$. Let G' be the subgraph of G with vertex set $V(H') \cap bd\Sigma$ and with edges those edges e of G with distinct ends u, v say, such that $H(u, v)$ is homotopic to e. Now reduction 1 does not apply, and hence G' is not connected. Let $(X_1 X_2)$ be a partition of $V(G')$ with X_1, $X_2 \neq \emptyset$, such that no edge of G' joins a vertex of X_1 to a vertex of X_2.

Let $i \in \{1, 2\}$. Suppose $a, b \in S_i \cap V(H')$ are distinct and consecutive in S_i, that is, $S_i(a, b)$ contains no vertex of H'. Since $V(H') \cap C_i = V(H) \cap S_i$, it follows that $S_i(a, b)$ contains no vertex of H, and so $S_i(a, b)$ is a segment with ends a, b.[F4] Then $H(a, b)$ is homotopic to $S_i(a, b)$ because S_i is a C_i-base for H'. Since (4.1)(i) does not apply, $P(a, b, S_i(a, b))$ exists; since reduction 2(ii) does not apply, every vertex of it is in $bd\Sigma$ and hence in $V(H)$; and since (4.1)(ii) does not apply, every vertex of it is in $V(H')$. We claim, moreover, that $P(a, b, S_i(a, b))$ is a subgraph of G'. For let e be an edge of $P(a, b, S_i(a, b))$ with ends u_1, u_2, and let Q_i be the subpath of $P(a, b, S_i(a, b))$ from u_i to b $(i = 1, 2)$. Since (4.1)(iii) does not apply, Q_i is homotopic to $H(u_i, b)$ $(i = 1, 2)$, and so $H(u_1, u_2)$ is homotopic to e. Thus $e \in E(G')$, and our claim is proved. Hence if $a \in X_j$ then $b \in X_j$ $(j = 1, 2)$. List the vertices of $V(H') \cap S_i$ in their order along S_i; each of them is consecutive to the next, and so all of them are in the same set X_j. It follows that for some j,

$$V(H') \cap S_i \subseteq X_j;$$

[F4] **Proof gap.** As printed, a, b are any two vertices of H' in S_i. But (4.1) and reduction 2 speak only of segments, and $S_i(a, b)$ is a segment only when a, b are consecutive. So the paragraph is proved for consecutive a, b, and two later steps (at the end of this paragraph, and on page 127) pass from consecutive pairs to all pairs. The statement of (4.2) holds as printed.

[p. 127] and since X_1, $X_2 \neq \emptyset$, we may renumber so that

$$V(H') \cap S_i = X_i \qquad (i = 1, 2).$$

Now H' is bad, and so we may assume that $|X_1| \geq 2$. Let K be the restriction of G' to X_1. For all ~~distinct~~ consecutive $a, b \in X_1$, $P(a, b, S_1(a, b))$ is a subgraph of K, and so K is connected, since the vertices of X_1, listed in their order along S_1, are each consecutive to the next.[F5] By (4.3) and the fact that S_1 is a C_1-base for H' there is a segment $S \subseteq S_1$ with ends s, t say which are adjacent in K. But then reduction 2 can be applied, a contradiction. This completes the proof of (4.2).

In view of (4.2) we see that (4.1) and reductions 1 and 2 serve to polynomially reduce the general ROUTED FOREST CONTAINMENT problem on the cylinder to its special case described in the conclusion of (4.2). That special case is the subject of the next section.

5. Disjoint Paths along a Cylinder

It is convenient in this section to take a concrete representation of Σ (using polar coordinates) as

$$\Sigma = \{(r, \theta) : 1 \leq r \leq 2, 0 \leq \theta < 2\pi\}$$

and of its cuffs C_1, C_2 as

$$C_i = \{(i, \theta) : 0 \leq \theta < 2\pi\} \qquad (i = 1, 2).$$

Let $f : [0, 1] \to \Sigma$ be continuous. Then it has a finite *winding number* $\theta(f)$, defined intuitively as $1/2\pi$ times the the total angle turned through (measured counterclockwise) by the line OX, where O is the origin, $X = f(x)$, and x ranges from 0 to 1. (This intuitive definition is sufficient for our purposes.) If P is a path drawn on Σ with one end in C_1 and the other in C_2, let $f : [0, 1] \to \Sigma$ be a continuous injection with image P and with $f(0) \in C_1$, $f(1) \in C_2$; then we define $\theta(P) = \theta(f)$. It is easy to see that this definition is independent of the choice of f. If P_1, P_2 are both paths drawn on Σ from some $s \in C_1$ to some $t \in C_2$, then $\theta(P_1) - \theta(P_2)$ is an integer, and is zero if and only if P_1 is homotopic to P_2.

Let $k > 0$ be some fixed integer, and let

$$M_i = \left\{ \left(i, \frac{2j}{k}\pi \right) : 1 \leq j \leq k \right\} \qquad (i = 1, 2).$$

[F5] **Proof gap.** The paragraph on page 126 proves that $P(a, b, S_1(a, b))$ is a subgraph of G' only for consecutive a, b, and this is enough for K to be connected.

Let $\mathcal{L}$ be a set of k paths drawn on Σ, pairwise disjoint, and each with an initial vertex in M_1 and a terminal vertex in M_2. We call such a set $\mathcal{L}$ a [p. 128] *linkage*. If $\mathcal{L}$ is a linkage then clearly $\theta(P)$ is constant for $P \in \mathcal{L}$, and we denote this common value by $\theta(\mathcal{L})$.

Throughout this section, G is a graph, properly drawn in Σ with

$$V(G) \cap bd(\Sigma) = M_1 \cup M_2.$$

A linkage in G is a linkage $\mathcal{L}$ every member of which is a subgraph of G. If θ is a real number, we say that θ is feasible in G if there is a linkage $\mathcal{L}$ in G with $\theta(\mathcal{L}) = \theta$. The problem we are concerned with in this section is, given a number θ and a graph G, is θ feasible in G? Evidently the results of Section 4 serve to reduce the general ROUTED FOREST CONTAINMENT problem on the cylinder to this problem.

The best approach to the problem of deciding if any given θ is feasible in G appears to be to consider the whole range of feasible values of θ. We shall see that they form an interval (meaning that if $\theta_1 \leq \theta_2 \leq \theta_3$ and θ_1, θ_3 are feasible then so is θ_2 provided that $k\theta_2$ is integral); and we are able to find the endpoints of the interval efficiently. But any given θ with $k\theta$ integral is feasible if and only if it lies between the endpoints, and so our original problem is solved. As a bonus the algorithm is easily adapted to construct a linkage $\mathcal{L}$ in G with $\theta(\mathcal{L}) = \theta$ when one exists.

Thus we turn to the following problem: given a linkage $\mathcal{L}$ in G, to determine if $\theta(\mathcal{L}) + 1/k$ is feasible in G. Let $\mathcal{L} = \{P_1, \ldots, P_k\}$, where for $1 \leq i \leq k$ the initial vertex of P_i is

$$s_i = \left(1, \frac{2i}{k}\pi\right).$$

For $1 \leq i \leq k$, let Σ_i be the component of

$$\Sigma - (P_1 \cup \cdots \cup P_k)$$

which contains the point

$$\left(1, \frac{2i+1}{k}\pi\right)$$

and let R_i be the region of G containing the same point. A separation of G is a pair (G_1, G_2) of subgraphs of G with $V(G_1) \cup V(G_2) = V(G)$, $E(G_1) \cup E(G_2) = E(G)$, and $E(G_1) \cap E(G_2) = \emptyset$. Throughout the following, subscripts should be read modulo k, so that P_0, for example, means P_k. We use the following four reductions.

Reduction 1. Delete any edge which for some i is incident with s_i and contained in Σ_{i-1}. When $k = 1$, the edge must moreover leave s_1 on the side of the strip that precedes s_1: in the universal cover of the annulus, some connected lift of the edge lies in the lift of the strip that precedes the chosen lift of s_1 and is incident with that lift of s_1.[F6]

[p. 129] *Reduction 2. If there is a separation (G_1, G_2) of G with $V(G_1) \cap V(G_2) \subseteq \{s_i\}$ for some i, and with $P_1, \dots, P_k$ all paths of $G_1 \neq G$, replace G by G_1.*

Reduction 3. If some s_i has valency 1 and its neighbour v is not in C_2, contract the edge joining s_i and v, moving v in Σ to s_i along the edge.

Reduction 4. If for some i there is a vertex v of P_i such that v is incident with R_i, but not every vertex and edge of P_i between s_i and v is incident with R_i, let Q_i be a path in G between s_i and v such that every vertex and edge of it is incident with R_i, and replace P_i by the concatenation of Q_i and the subpath of P_i between v and C_2. When $k = 1$, "incident with R_1" is read in the universal cover of the annulus: take the lift of R_1 that contains the lift of the gap following the chosen lift of s_1; the vertex v and the part of P_1 before it are tested on the chosen lift of P_1, and the lift of Q_1 that starts at the chosen lift of s_1 must reach the same lift of v through the closure of that lift of R_1.[F7]

Reduction 4 does not actually reduce the size of G; but it is easy to check that if $(G', \mathcal{L}')$ is produced by an application of reduction 4, then one of the other reductions can be applied to $(G', \mathcal{L}')$.

The basis of our algorithm is the claim that if $(G', \mathcal{L}')$ is produced from $(G, \mathcal{L})$ by applying one of these reductions, then $\theta(\mathcal{L}') + 1/k$ is feasible in G'

[F6] **Statement mistake.** For one path the complement of P_1 is a single strip, so the printed condition forgets on which side of P_1 the edge leaves s_1. Take two arcs P and B with the same ends, where B turns once around the annulus, and the linkage $\{P\}$. The printed Reduction 1 may delete B. Afterwards a radial helix misses the drawing and has potential 0, while before the deletion B is a linkage of winding 1, so by (5.2) every helix has potential at least 1. So (5.5) fails for the printed reduction, and so does (5.6), since winding 1 is feasible before the deletion and not after. The added condition keeps the side. For $k \geq 2$ it agrees with the printed one.

[F7] **Statement mistake.** For one path the printed condition loses which side of P_1 is meant, and (5.6) fails. Let s and t be joined by three edges: a radial edge P, an edge A close to it, and an edge B that turns once around the annulus, and let the linkage be $\{P\}$. The region R_1 meets all of A and B but no inner point of P, so Reduction 4 at $v = t$ may replace P by B. This changes the winding number from 0 to 1. The feasible winding numbers are exactly 0 and 1, so the next winding number is feasible before the step and not after it, contrary to (5.6). With the condition read in the universal cover, the lift of B from the chosen lift of s ends one turn away from the old contact, so the step is not allowed. For $k \geq 2$ this reading agrees with the printed one.

if and only if $\theta(\mathcal{L})+1/k$ is feasible in G. This will be proved as a consequence of a characterization, which will in turn be proved using the reductions.

We begin with the following.

(5.1) *Suppose that none of reductions 1,..., 4 apply to G. Then either some s_i has valency 1 and its neighbour is in C_2, or $\theta(\mathcal{L}) + 1/k$ is feasible.*

Proof. We may assume that no s_i has valency 1. But then for each i the boundary of R_i meets P_i in exactly the vertex s_i; and meets P_{i+1} in a vertex different from s_{i+1}. Thus there is a path Q_i of G with initial vertex s_i and terminal vertex t_i say, where t_i is a vertex of P_{i+1} different from s_{i+1}, with $Q_i \subseteq \Sigma_i \cup \{s_i, t_i\}$. Let P'_i be the concatenation of Q_i with the subpath of P_{i+1} between t_i and C_2. Then $\{P'_1, \ldots, P'_k\}$ is a linkage $\mathcal{L}'$ with

$$\theta(\mathcal{L}') = \theta(\mathcal{L}) + \frac{1}{k}.$$

Now we turn to our characterization. If $f : [0,1] \to \Sigma$ is continuous, we define

$$|f| = \{f(x) : 0 \le x \le 1\}.$$

We say f is *G-normal* if $e \cap |f| = \emptyset$ for every edge e of G; that is, every point of $|f|$ is either a vertex or lies in some region of G. If f is G-normal, we define $L(f) = L_G(f)$ to be

$$|\{x \in [0,1] : f(x) \in V(G)\}|.$$

[p. 130] Let

$$N_i = \left\{ \left(i, \frac{2j+1}{k}\pi \right) : 1 \le j \le k \right\} \qquad (i = 1, 2).$$

If $f : [0,1] \to \Sigma$ is continuous, G-normal, with $f(0) \in N_1, f(1) \in N_2$ and with $L_G(f)$ finite, we say f is a *helix* (in G). The following lemma is fundamental.

(5.2) *Let θ be feasible in G, and let f be a helix. Then $L(f) \ge k|\theta(f) - \theta|$, that is,*

$$\theta(f) - k^{-1}L(f) \le \theta \le \theta(f) + k^{-1}L(f).$$

The proof is clear. We abbreviate $\theta(f) + k^{-1}L(f)$ by $\phi(f)$ or $\phi_G(f)$. One form of our characterization is the following.

(5.3) *If θ is feasible in G and $\theta + 1/k$ is not, then there is a helix f with $\phi(f) = \theta$.* We prove this by means of the following two lemmas.

(5.4) *If $(G', \mathcal{L}')$ is produced from $(G, \mathcal{L})$ by one of the reductions 1,..., 4, and $\theta(\mathcal{L}') + 1/k$ is feasible in G', then $\theta(\mathcal{L}) + 1/k$ is feasible in G.*

Proof. This is trivial for reduction 4, and almost equally trivial for the other reductions. We give the details. Let $\mathcal{L}''$ be a linkage in G' with $\theta(\mathcal{L}'') = \theta(\mathcal{L}') + 1/k$. Suppose first that the reduction is 1, 2 or 4. Then G' is a subgraph of G, and so $\mathcal{L}''$ is a linkage in G. Moreover $\theta(\mathcal{L}') = \theta(\mathcal{L})$. For reductions 1 and 2 this holds since the linkage is not changed. For reduction 4 with $k \geq 2$ it holds since some P_j with $j \neq i$ is not changed, and $\theta(P)$ is the same for all members P of a linkage. For reduction 4 with $k = 1$ it holds since, by the reading of reduction 4 given above, the lift of the new path from the chosen lift of s_1 ends at the same point as the chosen lift of P_1. Hence $\theta(\mathcal{L}) + 1/k$ is feasible in G.

Now suppose that the reduction is 3, and let e be the edge joining s_i and v. Moving v to s_i along e is done by a continuous map π of Σ to itself, which maps the closure of e to s_i, is one-to-one elsewhere, and is joined to the identity by a deformation that fixes $bd(\Sigma)$. Each path of $\mathcal{L}'$ is drawn as the image under π of the corresponding path of $\mathcal{L}$, and so the two have the same winding number, since a deformation fixing $bd(\Sigma)$ does not change the winding number of a curve between points of $bd(\Sigma)$. Hence $\theta(\mathcal{L}') = \theta(\mathcal{L})$. The map π maps $V(G) - \{s_i\}$ one-to-one onto $V(G')$, with v mapped to s_i, and maps the edges of G other than e one-to-one onto the edges of G', preserving incidence. Read each path of $\mathcal{L}''$ back in G by this correspondence, and add s_i and e at the start of the path that now starts at v. Since e is the only edge of G at s_i, this gives a linkage $\mathcal{M}$ in G. Its image under π is $\mathcal{L}''$, and so $\theta(\mathcal{M}) = \theta(\mathcal{L}'') = \theta(\mathcal{L}) + 1/k$, by the same argument. F8

(5.5) *If $(G', \mathcal{L}')$ is produced from $(G, \mathcal{L})$ by one of the reductions 1,..., 4, and f' is a helix in G' then there is a helix f in G with $\phi_G(f) \leq \phi_{G'}(f')$.*

Proof. Again this is trivial for reduction 4, and quite easy for reductions 2 and 3. Let us deal with reduction 1. Suppose that e is an edge of G incident with s_i but contained in Σ_{i-1} for some i, and let G' be obtained from G by deleting e. Let f' be a helix in G'. If $|f'| \cap e = \emptyset$ we may take $f = f'$ and the theorem is true. If $|f'| \cap e \neq \emptyset$, let R' the region of G' in which edge e of G is drawn; let

$$X = \{x \in [0, 1] : f'(x) \in R'\}$$

so that $X \neq \emptyset$; and let $\bar{X}$ be the closure of X. Choose $z_2 \in \bar{X}$, maximum.

F8 **Missing proof.** The paper calls (5.4) trivial or almost trivial. The details above follow the proof in the formalization.

Then $f'(z_2) \in \bar{R}'$, and either $f'(z_2) \in N_2$ or $f'(z_2) \in V(G')$.

Now we may assume that (1) *There exists* $z_1 \in [0, z_2)$ *with* $f'(z_1) \in V(G')$.

[p. 131] For if not, then $f'(x) \in R'$ $(0 \leq x < z_2)$ and so

$$f'(0) = \left(1, \frac{2i-1}{k}\pi\right).$$

Then the concatenation, in the natural way, of the function

$$x \mapsto \left(1, \frac{2i+1-2x}{k}\pi\right) \qquad (0 \leq x \leq 1)$$

with f' gives another helix with $\phi_{G'}$ the same, which does satisfy (1). Choose $z_1 \in [0, z_2)$ with $f'(z_1) \in V(G')$ and with z_1 maximum. (This is possible by (1), for $L_{G'}(f')$ is finite, and so there are only finitely many values of z_1 satisfying (1).) Now R' is incident with s_i, and so there is a region R of G with $R \subseteq R'$, incident with s_i, and with $f'(z_2) \in \bar{R}$. Define $h\colon [0,1] \to \Sigma$ by

$$h(x) = \left(1, \frac{2i+1-x}{k}\pi\right) \qquad (0 \leq x \leq 1).$$

If $f'(z_2) = s_i$, define $f\colon [0,1] \to \Sigma$ by

$$\begin{aligned} f(x) &= h(2x) && (0 \leq x \leq \frac{1}{2}) \\ &= f'(2x - 1 + (2-2x)z_2) && (\frac{1}{2} \leq x \leq 1). \end{aligned}$$

If $f'(z_2) \neq s_i$, let $g\colon [0,1] \to \Sigma$ be continuous, with $g(0) = s_i$, $g(1) = f'(z_2)$ and $g(x) \in R$ $(0 < x < 1)$, and define $f\colon [0,1] \to \Sigma$ by

$$\begin{aligned} f(x) &= h(4x) && (0 \leq x \leq \frac{1}{4}) \\ &= g(4x-1) && (\frac{1}{4} \leq x \leq \frac{1}{2}) \\ &= f'(2x - 1 + (2-2x)z_2) && (\frac{1}{2} \leq x \leq 1). \end{aligned}$$

In either case, f is a helix in G. We shall show that $\phi_G(f) \leq \phi_{G'}(f')$. Let $f''\colon [0,1] \to \Sigma$ be defined by

$$\begin{aligned} f''(x) &= f'(2xz_2) && (0 \leq x \leq \frac{1}{2}) \\ &= f(1-x) && (\frac{1}{2} \leq x \leq 1). \end{aligned}$$

Then $\theta(f') = \theta(f) + \theta(f'')$ and $L_{G'}(f') = L_{G'}(f) + L_{G'}(f'') - l$, where

$$\begin{aligned} l &= 1 && \text{if } f'(z_2) = s_i \\ &= 2 && \text{if } f'(z_2) \notin V(G') \\ &= 3 && \text{if } f'(z_2) \in V(G') - \{s_i\}. \end{aligned}$$

[p. 132] But $L_{G'}(f) = L_G(f)$, and

$$\begin{aligned} \phi_{G'}(f') - \phi_G(f) &= \theta(f') - \theta(f) + k^{-1}(L_{G'}(f') - L_G(f)) \\ &= \theta(f'') + k^{-1}(L_{G'}(f'') - l) \end{aligned}$$

and so it suffices to show that $L_{G'}(f'') \geq l - k\theta(f'')$. Now

$$\begin{aligned} L_G(f'') = &|\{x \in [0, z_1) : f'(x) \in V(G')\}| \\ &+ |\{x \in [z_1, z_2) : f'(x) \in V(G')\}| \\ &+ |\{x \in [0, \frac{1}{2}] : f(x) \in V(G')\}|. \end{aligned}$$

But $f'(z_1)$ is incident with R', and in particular is contained in $\bar{\Sigma}_{i-1}$; thus,

$$|\{x \in [0, z_1) : f'(x) \in V(G')\}| \geq -k\theta(f'').$$

Second,

$$|\{x \in [z_1, z_2) : f'(x) \in V(G')\}| \geq 1$$

since $f'(z_1) \in V(G')$. And third,

$$\begin{aligned} |\{x \in [0, \frac{1}{2}] : f(x) \in V(G')\}| &= 1 && \text{if } f'(z_2) \notin V(G') - \{s_i\} \\ &= 2 && \text{if } f'(z_2) \in V(G') - \{s_i\} \end{aligned}$$

and so in either case is at least $l - 1$. Adding, the result follows. Now we can prove the characterization.

Proof of (5.3). We proceed by induction on $|V(G)| + |E(G)|$. If reduction 1, 2 or 3 applies to $(G, \mathscr{L})$ then the result follows from our inductive hypothesis and (5.4), (5.5). If reduction 4 applies and results in $(G', \mathscr{L}')$ say, then as we observed before one of reductions 1, 2 or 3 applies to $(G', \mathscr{L}')$ and hence the theorem is true for $(G', \mathscr{L}')$; and again by (5.4), (5.5) it is true for $(G, \mathscr{L})$. If none of reductions 1,..., 4 applies, then by (5.1) some s_i has valency 1 and its neighbor t say is in C_2, since $\theta(\mathscr{L}) + 1/k$ is not

feasible. Let e be the edge incident with s_i; then P_i consists solely of the edge e. Let $g\colon [0,1] \to \Sigma$ be continuous, with

$$g(0) = \left(1, \frac{2i+1}{k}\pi\right), \qquad g(1) = t,$$
$$g(x) \in R_i \qquad (0 \le x < 1).$$

Now

$$t = \left(2, \left(\frac{2i}{k} + 2\theta(\mathscr{L})\right)\pi\right).$$

[p. 133] Let $h\colon [0,1] \to \Sigma$ be defined by

$$h(x) = \left(2, \left(\frac{2i-x}{k} + 2\theta(\mathscr{L})\right)\pi\right) \qquad (0 \le x \le 1).$$

Let $f\colon [0,1] \to \Sigma$ be defined by

$$\begin{aligned} f(x) &= g(2x) && (0 \le x \le \frac{1}{2}) \\ &= h(2x-1) && (\frac{1}{2} \le x \le 1). \end{aligned}$$

Then f is a helix, $L(f) = 1$ and $\theta(f) = \theta(\mathscr{L}) - 1/k$. Hence $\phi(f) = \theta(\mathscr{L})$, as required. As a corollary we have the following justification for our algorithm.

(5.6) *If $(G', \mathscr{L}')$ is produced from $(G, \mathscr{L})$ by one of the reductions 1,..., 4, then $\theta(\mathscr{L}') + 1/k$ is feasible in G' if and only if $\theta(\mathscr{L}) + 1/k$ is feasible in G.*

Proof. "Only if" follows from (5.4). To prove "if", suppose that $\theta(\mathscr{L}') + 1/k$ is not feasible in G'. Then by (5.3) there is a helix f' in G' with $\phi(f') = \theta(\mathscr{L}')$, and by (5.5) there is a helix f in G with $\phi(f) \le \phi(f')$. But $\theta(\mathscr{L}') = \theta(\mathscr{L})$, and so

$$\phi(f) \le \phi(f') = \theta(\mathscr{L}') = \theta(\mathscr{L}).$$

Hence by (5.2), $\theta(\mathscr{L}) + 1/k$ is not feasible in G. Let us state the algorithm.

(5.7) *Algorithm to decide if $\theta(\mathscr{L}) + 1/k$ is feasible.* We apply any of the reductions which can be applied, transforming to smaller equivalent problems by (5.6). If none of the reductions apply we determine feasibility by (5.1).

The algorithm can easily be adapted to provide us with a linkage $\mathscr{L}'$ satisfying $\theta(\mathscr{L}') = \theta(\mathscr{L}) + 1/k$ if one exists. Its running time is polynomially

bounded, as may easily be checked. We check this. We apply reduction 4 only when none of reductions 1, 2, 3 applies. Each of reductions 1, 2, 3 makes $|V(G)| + |E(G)|$ smaller, and reduction 4 does not change G. After an application of reduction 4 one of the other reductions applies, as we observed after the list of reductions, and so no two successive steps both use reduction 4. Let $b = 0$ if the last step used reduction 4, and $b = 1$ otherwise (in particular before the first step). A step by reduction 1, 2 or 3 decreases $|V(G)| + |E(G)|$ by at least 1 and increases b by at most 1, and a step by reduction 4 decreases b from 1 to 0. Hence each step decreases $2(|V(G)| + |E(G)|) + b$ by at least 1. This number is $2(|V(G)| + |E(G)|) + 1$ for the original graph, and at least 1 at the end, since the last step does not use reduction 4. So the number of steps is at most twice $|V(G)| + |E(G)|$ for the original graph. Each step takes polynomial time. The regions of G, the strips Σ_i, and the vertices and edges incident with each R_i may be read from the drawing, and with them reductions 1, 2 and 3 are tested directly. For reduction 4, a vertex v and a path Q_i may be found by a search from s_i in the subgraph of the vertices and edges incident with R_i, read on the side of P_i that reduction 4 refers to. When no reduction applies, (5.1) asks only whether some s_i has valency 1 and a neighbour in C_2. To find a linkage $\mathcal{L}'$ with $\theta(\mathcal{L}') = \theta(\mathcal{L}) + 1/k$, at the end we build the paths Q_i of the proof of (5.1) by following the boundary of R_i from s_i until it first meets P_{i+1}. Then we undo the steps in reverse order. Deletions need no change, since a linkage of a subgraph is a linkage of the graph, and a contraction by reduction 3 is undone by adding the contracted edge at the start of the path that begins at its end, as in the proof of (5.4).[F9]

As further corollaries we have

(5.8) *If θ is feasible in G, the following are equivalent:* (i) *there is a helix f in G with $\phi(f) = \theta$* (ii) *$\theta + 1/k$ is not feasible* (iii) *no $\theta' > \theta$ is feasible.*

Proof. By (5.2), (i) $\Rightarrow$ (iii). Trivially (iii) $\Rightarrow$ (ii), and by (5.3), (ii) $\Rightarrow$ (i).

[p. 134] (5.9) *If $\theta_1 \le \theta_2 \le \theta_3$ and θ_1, θ_3 are feasible in G and $k\theta_2$ is an integer, then θ_2 is feasible in G.*

This follows easily from (5.8). Now we apply these results to the main problem of this section, that of deciding if any given θ is feasible in G. We have the following characterization. (It is indeed a characterization, because Menger's theorem tells us when (5.10)(ii) is true.)

[F9] **Missing proof.** The paper says that the running time may easily be checked. The count above, with the measure $2(|V(G)| + |E(G)|) + b$, is the one used in the formalization.

(5.10) *θ is feasible in G if and only if* (i) *$k\theta$ is an integer;* (ii) *there is some linkage in G; and* (iii) *for every helix f, $\theta(f) - k^{-1}L(f) \le \theta \le \theta(f) + k^{-1}L(f)$.*

Proof. Necessity is clear. For sufficiency, let θ_1 be maximum and θ_2 minimum such that θ_1, θ_2 are feasible. (This is possible by (ii).) Then there is a helix f_1 in G with

$$\theta(f_1) + k^{-1}L(f_1) = \theta_1$$

by (5.3); and similarly (reversing the orientation of the plane) there is a helix f_2 with

$$\theta(f_2) - k^{-1}L(f_2) = \theta_2.$$

Hence by (iii), $\theta_2 \le \theta \le \theta_1$, and so by (i) and (5.9), θ is feasible in G. For an algorithm, we have

(5.11) *Algorithm to decide if θ is feasible in G.* We check that conditions (i) and (ii) of (5.10) hold; if not then θ is not feasible. Checking (ii) is simply a maximum-flow problem and so can be done efficiently. We assume that we find a linkage $\mathscr{L}$, and (by reversing the orientation of the plane if necessary) that $\theta(\mathscr{L}) \le \theta$. If $\theta(\mathscr{L}) = \theta$ we are finished. If not, we use (5.7) to test if $\theta(\mathscr{L}) + 1/k$ is feasible, and if so to find a linkage $\mathscr{L}'$ with $\theta(\mathscr{L}') = \theta(\mathscr{L}) + 1/k$. (If there is no such linkage then θ is not feasible.) If we find $\mathscr{L}'$, we replace $\mathscr{L}$ by $\mathscr{L}'$ and continue. At each replacement the magnitude of $\theta - \theta(\mathscr{L})$ is reduced by $1/k$, and so the algorithm terminates.

To estimate the running time, we need to bound the number of applications of (5.7). This is bounded by

$$1 + \max |k(\theta_1 - \theta_2)|,$$

[p. 135] the maximum being taken over all θ_1, θ_2 which are both feasible in G. In turn, we can bound that with the following theorem.

(5.12) *If θ_1, θ_2 are both feasible in G then*

$$|\theta_1 - \theta_2| \le \frac{1}{k^2}|V(G)|.$$

Proof. Let $\mathcal{L}_i$ be a linkage with $\theta(\mathcal{L}_i) = \theta_i$ $(i = 1, 2)$. For each $P \in \mathcal{L}_2$ there is a helix f in G with

$$|f| \cap V(G) = V(P)$$
$$\theta(f) = \theta(P)$$

as is easily seen. By (5.2),

$$L(f) \geq k|\theta(f) - \theta(\mathcal{L}_1)|$$

and so

$$|V(P)| \geq k|\theta_2 - \theta_1|.$$

But distinct P's in $\mathcal{L}_2$ are vertex-disjoint; and so

$$|V(G)| \geq k^2|\theta_2 - \theta_1|$$

as required.

It follows that the running time of the algorithm (5.11) is polynomially bounded. This, and the methods described in the last section, evidently yield a polynomially bounded algorithm for the general ROUTED FOREST CONTAINMENT problem on the cylinder.

We mention that for ROUTED FOREST CONTAINMENT on the disc we gave an algorithm and a structural characterization, while for the problem on the cylinder we have only given an algorithm for the general case and a characterization for the special case treated in this section. It is possible to find a structural characterization for the general problem on the cylinder by combining conditions like those of (3.6)(ii) with those of (5.10)(ii), (iii), but we omit the details.

6. Conclusion

We thus have a polynomially bounded algorithm to solve ROUTED FOREST CONTAINMENT on the cylinder. Let us see how to use it to solve DISJOINT CONNECTING SUBGRAPHS on the cylinder.

[p. 136] We are given a cylinder Σ, a graph G properly drawn on Σ, and a division Δ with

$$\bigcup \Delta = V(G) \cap bd(\Sigma).$$

We first check that some graph drawn on Σ realizes Δ (this is akin to checking the "cross-free" condition of (3.4) and can be done efficiently be adapting a planarity testing algorithm). Any realization of Δ is a forest with a homotopy class, and given the homotopy class we can test for the existence of a subgraph of G in this class using ROUTED FOREST CONTAINMENT. Thus we are finished provided that we can satisfactorily bound the number of different homotopy classes to be checked.

If $s, t \in V(H) \cap bd\Sigma$ lie in the same component of H, let $H(s,t)$ denote the path of the forest H between them, as before. Let C_1, C_2 be the cuffs of Σ. To specify the homotopy class of H, it suffices to specify

(i) the homotopy class of $H(s,t)$ for every pair of vertices $s,t \in V(H) \cap C_i$ for which $H(s,t)$ exists $(i = 1, 2)$, and

(ii) if some $\delta \in \Delta$ meets both C_1 and C_2, the homotopy class of $H(s,t)$ for some $s \in V(H) \cap C_1$ and $t \in V(H) \cap C_2$ for which $H(s,t)$ exists,

as is easily seen. Let Σ' be the disc obtained from Σ by "sewing" a disc onto C_2. The number of possibilities involved in (i) when $i = 1$ is at most the number of regions of a forest F properly drawn in Σ' with

$$V(F) \cap bd(\Sigma') = \bigcup \Delta \cap C_1;$$

that is, it is at most $|\bigcup \Delta \cap C_1|$. Similarly when $i = 2$ the number of possibilities is at most $|\bigcup \Delta \cap C_2|$. Thus the total number of possible assignments in (i) is at most

$$\left|\bigcup \Delta \cap C_1\right| \cdot \left|\bigcup \Delta \cap C_2\right|.$$

To bound the number of assignments we need consider in (ii) we proceed as follows. Choose $s \in C_1 \cap V(H)$ and $t \in C_2 \cap V(H)$ such that $H(s,t)$ exists. (If there are no such s,t then (ii) is vacuous and presents no problem.) We need only consider homotopy classes for $H(s,t)$ which are feasible in G. By (5.12) with $k = 1$, there are at most $|V(G)|$ of these, and we can efficiently find them by using the algorithm (5.11). Thus the total number of homotopy classes for H that we need to check is at most

$$\left|\bigcup \Delta \cap C_1\right| \cdot \left|\bigcup \Delta \cap C_2\right| \cdot |V(G)|.$$

[p. 137] Moreover we can find all these homotopy classes efficiently, and so we have a polynomially bounded algorithm for DISJOINT CONNECTING SUBGRAPHS on the cylinder.

One can do much better than this. The algorithm above was given for its simplicity, but we sketch another approach. Once again we have at most

$$\left|\bigcup \Delta \cap C_1\right| \cdot \left|\bigcup \Delta \cap C_2\right|$$

possibilities for the homotopy classes of the $H(s,t)$'s where s,t are in the same cuff of Σ. Having chosen one of the possibilities to check, we can now apply reduction arguments like those in section 4 to reduce to the case where each $\delta \in \Delta$ has

$$|\delta \cap C_1| = |\delta \cap C_2| = 1.$$

This case can be solved easily with at most $|\bigcup \Delta \cap C_1|$ applications of (5.7).

So far we have considered our problem DISJOINT CONNECTING SUBGRAPHS on a disc and on a cylinder. What about other surfaces? As mentioned in the introduction, it will be shown in [3] that there is a polynomially bounded algorithm to solve this problem on any fixed surface provided that $|\bigcup \Delta|$ is kept bounded. This is not a complete generalization of the results of this paper, because

(i) in this paper we do not need to keep $|\bigcup \Delta|$ bounded, and

(ii) we have no structural characterization in general.

However, a full analogue does exist when Σ is the Möbius band. Working with ROUTED FOREST CONTAINMENT, one reduces (using the methods of section 4) to the case when each component of the forest H is a path with only its end-vertices in $bd(\Sigma)$, and none of these paths is homotopic to a portion of $bd(\Sigma)$. In that case, a necessary and sufficient condition for G to contain a subgraph homotopic to H is that there is no closed sequence

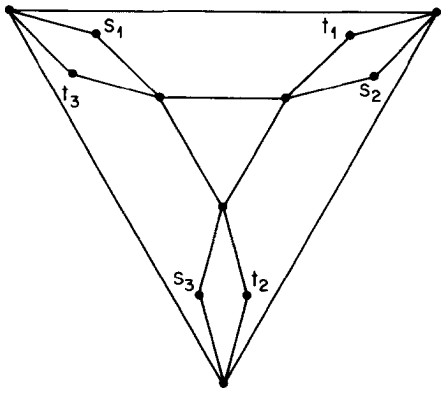
$$v_0, R_1, v_1, R_2, \ldots, v_{k-1}, R_k, v_k = v_0$$

of vertices and regions of G (in which consecutive members are incident) which is "homotopic" to $bd(\Sigma)$ and with

$$k < |bd(\Sigma) \cap V(G)|.$$

It is possible to base a polynomially bounded algorithm on that fact, but we omit the details.

Let G be a planar graph, and let $s_1, t_1, \ldots, s_k, t_k$ be distinct vertices of G,



[p. 138]

FIGURE 1

constrained to lie on some bounded number ρ of regions of G. We have given a way to solve the Disjoint Connecting Paths problem for this config-

uration when $\rho = 1$ and when $\rho = 2$. An example due to L. Lovász indicates that the $\rho = 3$ case is less tractable (at least from the point of view of polyhedral combinatorics). When $\rho = 3$, one can sometimes do better with a "half-integral" packing of paths than with an integral one. (It follows easily from the results of this paper that such a phenomenon does not occur when $\rho = 1$ or 2.) In the graph of figure 1, there do not exist three mutually disjoint paths joining s_i and t_i $(i = 1, 2, 3)$, respectively; but there do exist six paths, using each vertex at most twice, such that for $i = 1, 2, 3$, two of them join s_i and t_i (Fig. 1).

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